Settings and results: the supplement, run J and the published model

Item	Supplement (16 runs)	Run J waypoints (24 runs)	Davey et al. (2016)
Seeds	1001–1016 (16)	1–24 (24)	—
Particles per run	7,000,000	7,000,000	—
Particles per mode (TH, MH, TT, MT, LN)	1.2, 1.3, 1.1, 2.2, 1.2 M	1.0, 0.5, 2.5, 0.5, 2.5 M	—
00:11 hand-off rows per run	100,000 normal + 100,000 supplement	100,000	—
Supplement tilt	r = 0.1 + 0.9 h/H; 45/40/15 % over 34.5–29.5 / 29.5–25.5 / 25.5–21.0°S	none	—
BFO noise sd	4.3177 Hz	4.3177 Hz	4.3177 Hz (Table 5.1)
18:25:27 R600 burst	held out	held out	used
Fuel-flow factor on extrapolated steps	N(1.0, 0.046), provisional	N(1.0, 0.046), provisional	no fuel model
Engine binary	3bf0801 (run J + flag)	1244b7c (tag run-j)	—
Median latitude at 00:19	37.11°S	37.10°S	37.43°S
95 % interval at 00:19	33.25–38.02°S	33.05–37.65°S	34.83–39.21°S
P(34.5–36.5°S) at 00:19	0.140	0.167	0.270
P(north of 33°S) at 00:19	0.0232	0.0246	0.0040
ln Z (pooled)	-127.77	−127.82 (seed sd 0.26)	—
Split-half, mean of balanced partitions	0.883 (16 runs); 8-run subsets 0.837	0.934 (24 runs); 8-run subsets 0.892	—

Plain: The table compares the settings and the results of the supplement runs with run J and with the published model. The supplement uses the run J model; only the particle split between the autopilot modes and the extra 00:11 hand-off are different. The 00:19 rows use the final result, with both 00:19 BTO values and no 00:19 BFO. They agree with run J within the run-to-run spread. The source is the core filter: run J, early-flight family 'waypoints' (24 runs), and the northern supplement (16 new runs). The prior track starts from the 18:01 radar position and follows a random route through the 2014 waypoints. The 00:11 data are used; the 00:19 data are not. The one-engine physics and the fuel-flow factor are provisional. No search areas, ocean drift or end-of-flight data are applied. The 18:25:27 signal is not used. The other three early-flight families are not included.

Technical: Supplement 00:19 values from the pooled summary.json of the 16 runs (mh370 summarise over merged seeds; evidence pooling by mode). Run J values from results/run-j/README.md (24 runs). Davey et al. (2016), Fig. 10.3: vector-recovered curve (engine/.sources/davey-2016/fig10-3-bto-bfo-latitude-pdf-vector.csv, 5315fde); its quantiles and band masses computed here on a 0.01 deg grid. Split-half: mean over all balanced partitions (6,435 for 16 runs); no calibrated floor for 16 runs; run J's ruling (PASS, Pete 16 Oct 2026) is not affected. Southern 2.5 % point differs (38.02 against 37.65°S): a run-to-run effect of the 38°S side mode (Monte Carlo noise; see the standard report, Figure 1). Core 00:11 hand-offs (filtering distribution at m0011, look-ahead correction exp(log_correction) applied): run J waypoints seeds 1-24 (tag run-j, 1244b7c; 100k rows per seed) and supplement seeds 1001-1016 (rev 3bf0801; particles per mode [TH, MH, TT, MT, LN] = [1.2, 1.3, 1.1, 2.2, 1.2]M vs run J [1.0, 0.5, 2.5, 0.5, 2.5]M; supplement hand-off 100k rows per seed, tilt r(lat) = 0.1 + 0.9 h/H over 00:11 latitude -34.5..-21.0). BFO sd 4.3177 Hz; kappa_x ~ N(1.0, 0.046) and one-engine s7 + s8 PROVISIONAL; 18:25:27 R600 held out. Search case: none; ocean models: none; end-of-flight 00:19 options: not applied. Combination rule B: balance heuristic with r_hat (r_hat.csv), n1 = 2,399,999, n2 = 1,600,003 rows.